

Beyond the Badge: Designing Digital Mental Health Interventions for Korean Investigative Officers

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Abstract

This study explores the unique mental health challenges faced by Korean investigative police officers and how these shape their help-seeking behaviors within cultural and occupational contexts. Based on interviews with 19 officers, four key themes emerged: hierarchical organizational stress, internalization of a responsible police image, routine exposure to negative events, and misalignment between internal growth needs and external acknowledgement. The findings provide actionable design implications for tailored digital mental health interventions, such as integrating micro-interventions into workflows to address hierarchical organizational stress and demanding workloads, recognizing emotional labor through positive rewards to support the moral and self-sacrificing police identity, and fostering mutual reliance via anonymous digital communication platforms to mitigate the effects of routine exposure to negative events. These strategies address officers' specific challenges while aligning with their professional and cultural nuances. Although focused on Korean investigative police officers, the study offers valuable insights for designing role-specific and culturally informed solutions in other high-stress professions.

CCS Concepts

• **Human-centered computing** → **User centered design**; *Contextual design*.

Keywords

digital mental health intervention, occupational mental health intervention, culturally informed design, investigative police officer

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1 Introduction

Designing digital mental health interventions (DMHIs) for specific occupational groups is crucial for addressing their unique needs and challenges. Occupational contexts often shape not only the types of stressors individuals face but also their attitudes toward help-seeking and mental health care[14, 18]. Prior research illustrates the value of tailoring interventions to meet these specialized demands, resulting in improved adoption and effectiveness[13, 18, 22, 27]. For example, mobile health technologies for military personnel have been specifically designed to address challenges encountered during deployment, incorporating features such as remote counseling and PTSD-specific resources[28]. Similarly, a web-based intervention for wildland firefighters has integrated physical resilience training, stress management, and nutrition-focused modules to mitigate the adverse effects of extended fire seasons and physically demanding work[18]. In healthcare, mindfulness programs and peer-support networks for physicians have shown to significantly reduce depression and anxiety symptoms, while organizational interventions have targeted systemic stressors such as excessive work hours and poor communication[22]. These examples highlight how grounding DMHIs in the occupational demands and lived experiences of the target population can lead to more impactful and sustainable outcomes.

Police officers, recognized globally as a high-risk group for mental health challenges[2, 9, 12, 14, 26], exemplify the need for such tailored interventions. Policing is consistently ranked as one of the most stressful occupations worldwide due to frequent exposure to trauma, shift-based work, and the high-pressure nature of their responsibilities[14]. While some initiatives exist—such as physical health management apps and offline services—they often lack

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the context-specific nuance required for sustained adoption and impact[8, 21, 23, 32].

Understanding the mental health challenges of police officers requires a more granular approach that goes beyond the general category of "policing" to account for the differences in roles and contexts. Police officers' job responsibilities vary significantly across departments, resulting in vastly different sources and levels of stress[9]. Among the various divisions, investigative police officers in South Korea face unique stressors shaped by their occupational demands[16, 19, 20]. These officers primarily handle cases such as fraud, embezzlement, and election-related offenses, which are distinct from the violent crimes typically managed by criminal investigation divisions. While direct visits to crime scenes are less frequent, investigative officers often manage suspects and wanted individuals in detention, resulting in frequent exposure to offenders. Additionally, they are tasked with resolving civil complaints, processing legal filings, and mediating between suspects, victims, and complainants. Following the 2023 investigative rights reform, these officers have experienced an increased administrative burden, including case reception, report submissions, and transfer processing. This combination of diverse and emotionally charged responsibilities underscores the necessity of developing mental health interventions tailored to the specific needs of investigative officers.

The cultural context in South Korea adds another layer of complexity to the challenges faced by police officers. In this collectivist society, which emphasizes interdependence and harmony, societal expectations and heightened public scrutiny place unique pressures on officers[1]. According to previous research, Korean police officers often face more criticism from the public compared to their counterparts in other countries, further intensifying these challenges[25]. These dynamics may contribute to distinct influences on their mental health, potentially shaping their attitudes toward seeking help and their overall well-being. This study represents an initial step toward designing culturally and contextually tailored DMHIs for Korean investigative police officers. Through a phenomenological exploration of their mental health challenges and help-seeking behaviors, we aim to identify meaningful intervention touchpoints. Semi-structured interviews with 19 investigative officers from diverse career backgrounds were conducted, providing insights into their lived experiences. The study seeks to answer three primary research questions:

- What unique mental health challenges arise within the occupational contexts of Korean investigative police officers?
- How do the unique occupational contexts of Korean investigative police officers manifest in their help-seeking attitudes and behaviors?
- What are the design implications for digital mental health services tailored to the unique needs of Korean investigative police officers?

By addressing these questions, this study contributes to the development of DMHIs that are not only role-specific but also culturally grounded, offering a roadmap for supporting the well-being of investigative police officers and other high-stress occupational groups.

2 Method

2.1 Participants and Recruitment

We conducted semi-structured interviews with nineteen metropolitan police officers working in the investigative department of a single police station in South Korea. The recruitment process was initiated by our research team, which formally requested participation through the Korean National Police Agency. Following this request, an official notice was distributed to investigative divisions, requesting participation in a brief psychological consultation and interview on mental well-being. Officers who volunteered in response to this notice took part in the study.

A summary of participants' demographic characteristics, including age, investigative role experience, and prior mental health service experience, is presented in *Table 1*. Participants varied in age and years of experience in investigative roles, ranging from junior officers with fewer than five years of service to senior investigators with over twenty years of experience. To assess their current mental health status, we incorporated Patient Health Questionnaire-9 (PHQ-9) scores into the demographic data (Mean = 6.26, SD = 5.09). These scores were used as contextual information to provide insight into officers' mental health conditions rather than for comparative analysis.

Regarding prior engagement with mental health services, only one participant had previously utilized police-provided mental health counseling services, while another had sought psychiatric consultation from an external institution. The remaining seventeen officers had never accessed any mental health services.

2.2 Data Collection

This study received approval from the *UNIST Institutional Review Board* (IRB Number: UNISTIRB-24-042-A).

We conducted in-person, one-on-one interviews at the participants' workplace, ensuring a private and confidential setting. A designated private room, separate from their workspace, was secured within the police station for interviews. This setup allowed participants to speak openly without concern for workplace visibility while maintaining confidentiality. Conducting interviews in their professional environment also enabled the researcher to observe officers' workspaces, providing additional contextual insights incorporated into the analysis.

Interview Protocol. A semi-structured interview guide was developed based on the study objectives, focusing on three main areas: (1) mental health challenges specific to investigative police officers, (2) occupational and organizational stressors, and (3) attitudes and barriers to seeking mental health support. Before conducting the study interviews, we refined the guide through two pilot interviews—one with a police mental health counselor and another with an investigative officer. These pilot sessions helped ensure that the questions effectively captured the lived experiences of the participants and were appropriately structured for the study.

Interview Procedure. Interviews lasted between 13 to 53 minutes (average = 24 minutes). With verbal consent, all interviews were audio-recorded and transcribed verbatim using Naver Clova. Transcriptions were manually reviewed against the original recordings to ensure accuracy. Selected quotes were translated from Korean to English by the first author for inclusion in this study.

Table 1: Demographic Information of Participants.

Participant ID	Gender	Length of Experience	PHQ-9 Score (Severity Level)	Mental Health Service Experience
P1	Male	2 years	18 (Moderately Severe)	Private Mental Clinic (Offline)
P2	Female	2 years	15 (Moderately Severe)	None
P3	Male	20 years	9 (Mild)	None
P4	Male	8 years	8 (Mild)	None
P5	Female	1 year	7 (Mild)	None
P6	Male	10 years	8 (Mild)	None
P7	Male	27 years	7 (Mild)	None
P8	Female	3 years	6 (Mild)	None
P9	Female	6 months	2 (None)	None
P10	Female	9 months	3 (None)	None
P11	Male	7 years	3 (None)	None
P12	Female	-	2 (None)	None
P13	Male	1 year	3 (None)	None
P14	Female	2 months	15 (Moderately Severe)	None
P15	Male	9 months	2 (None)	None
P16	Male	3 years	7 (Mild)	None
P17	Male	2 years	2 (None)	None
P18	Female	2 years	1 (None)	None
P19	Male	4 years	1 (None)	Police Counseling Service (Offline)

The "Length of Experience" column represents the number of years participants have worked in investigative roles. The "Mental Health Service Experience" column refers to the experience of any mental health service. *Police Counseling Service* is a service provided by the National Police Agency for free, and *Private Mental Clinic* is a external service that they need to use at their own expense. Participant P12 chose not to disclose their length of experience.

2.3 Ethical Considerations

Given the sensitive nature of the discussions, we implemented several ethical precautions to safeguard participant well-being.

Pre-Interview Psychological Support. Before each interview, participants attended a 30-minute psychological counseling session conducted by a professional mental health counselor. During this session, officers were provided with a safe space to openly discuss any psychological distress they were experiencing. The counselor also provided guidance on potential follow-up support options, ensuring that participants were aware of available mental health resources before engaging in the study.

Confidentiality and Psychological Safety. All interviews took place in a private setting within the police station, eliminating concerns about workplace visibility and allowing officers to speak candidly. If participants expressed distress during the interviews, they were provided with information on available mental health support resources. No participants required immediate intervention at the time of the study.

Post-Interview Well-being Monitoring. After the interviews, participants were given the opportunity to express any discomfort or request follow-up support. Although no immediate interventions were needed, officers were informed that they could seek additional support through the police mental health system.

Data Protection and Anonymity. To ensure confidentiality, all personally identifiable information was removed from the transcripts. Participants were assigned anonymous identifiers, and all data were stored securely in accordance with institutional research ethics guidelines.

2.4 Data Analysis

The interviews were analyzed using a reflexive thematic analysis approach[5], with a phenomenological approach to deeply explore the lived experiences of participants[3, 17]. The analysis was conducted by the primary researcher who also conducted the interviews, alongside two additional researchers who participated in the project design and were familiar with the context of the study.

The researchers first familiarized themselves with the data by extensively reading the interview several times. Using Excel, initial codes were applied to summarize key ideas. Expanded codes were subsequently developed, incorporating the surrounding context of participants' narratives to provide a more nuanced understanding of the data. Based on the expanded codes, the first author categorized and grouped related codes into potential themes and subthemes using Miro. This process involved iterative discussions with four co-authors to ensure alignment with the research questions and to refine the emerging themes. To uncover deeper insights, an iterative process was undertaken to re-evaluate and refine the themes. The analysis aimed to identify themes that directly addressed research questions, focusing on uncovering meaningful patterns in the data. This process resulted in four themes that were robustly linked to the participants' experiences and provided actionable insights for the study's objectives. The reflexive thematic analysis approach allowed the research team to go beyond surface-level observations and uncover patterns in the data that aligned with the occupational and cultural contexts of investigative police officers. This iterative process facilitated the identification of actionable themes and subthemes, which are presented in the findings.

3 Findings

3.1 Hierarchical Organizational Stress

The hierarchical culture of Korean police organizations creates a unique blend of organizational and operational stress for officers. Rigid structures and ingrained norms limit feedback opportunities and intensify workload pressure. These factors combine to create a work environment that profoundly impacts officers' mental health and professional motivation.

3.1.1 Rigid Organizational Structures and Learned Helplessness.

The rigid nature of the police hierarchy fosters a culture of learned helplessness among officers. The deeply entrenched "top-down" command system stifles constructive feedback and discourages efforts to advocate for change. Officers perceive those who voice concerns during meetings as naive or unaware of organizational norms. One officer lamented:

"I used to be more optimistic and felt a sense of pride and excitement about the police motto, but now I've become more skeptical."(P4)

Officers internalize this mistrust toward the organization, leading them to believe that decisions, such as personnel changes, are made without regard for their welfare or perspectives.

3.1.2 Demanding and Unpredictable Work Characteristics. The unpredictable nature of police work, combined with relentless pressure to meet tight deadlines, significantly amplifies stress levels. Officers are often tasked with responsibilities that naturally take time to resolve, yet they face constant demands from superiors, creating a sense of inadequacy and anxiety:

"Deadlines must be met for resolving cases, but if the complainant makes excuses and doesn't show up, the timeline gets delayed. ... Meanwhile, my superior keeps pressuring me to finish quickly, and that's when I started experiencing anxiety symptoms."(P7)

This cycle of unrealistic demands and inherent delays imposes constant pressure, leaving many officers feeling trapped. Such unrelenting stress leads to severe mental and emotional exhaustion, sometimes to the extent that they can understand why some colleagues resort to taking their own lives under similar pressure. Furthermore, these circumstances foster a profound sense of helplessness, as officers perceive themselves as scapegoats in a system that prioritizes shifting blame over supporting its members:

"When someone dies, someone has to be held accountable. In those situations, it's the commanding officers who take responsibility for the duties performed."(P3)

The challenge is further compounded by a 'work over well-being' mindset, where officers neglect their mental health even in the face of severe distress. For example, some officers reported taking no action to address their mental health struggles until they reached the brink of suicidal ideation. The constant influx of unpredictable tasks exacerbates the problem, leaving officers feeling unable to schedule time for mental health programs. As one participant noted:

"If I have to attend a mental health-related program and complete it, I can't go if the cases pile up during that time."(P14)

Similarly, another officer expressed how the pressures of their workload leave little room for digital mental health interventions, even if readily available. This recurring tension between occupational demands and well-being highlights a critical gap: the absence of formalized mental health support within police work schedules. Many officers emphasized the need for mental health services to be 'officially integrated' into their routines, ensuring that participation becomes mandatory rather than optional. One officer explained:

"I think it would be better if it became mandatory on a set day—whether it's counseling or a health checkup, something we are required to take time for and actually go do."(P12)

3.2 Internalization of a Responsible Police Image

The sense of duty and moral responsibility central to police work is undoubtedly a vital virtue, but it paradoxically serves as a significant source of mental strain. For Korean investigative officers, their deep commitment to helping others and upholding justice often leads to heightened emotional burdens and self-sacrifice, which can conflict with the realities of overwhelming workloads and systemic inefficiencies.

3.2.1 Deep Commitment to Investigative Duties. Korean police officers often view their profession as a moral obligation to assist those in need. This belief fosters a profound sense of responsibility. In investigative work, this responsibility becomes magnified due to the complexity of cases, which often involve multiple stakeholders and significant consequences for victims. Officers are acutely aware that their decisions can directly impact lives, sometimes to the extent of life or death. This pressure becomes even more intense when officers handle high-stakes cases involving numerous victims. Even during leisure time, officers struggle to disconnect from work. This inability to separate work from personal life often results in ongoing stress, and some have reported experiencing physical symptoms, such as heart palpitations and chronic insomnia.

"For us, it's just work, but for them, it could be a matter of life and death... When the case is large and involves many victims... taking on such an important case makes it hard to sleep, and even when I close my eyes, it's all I can think about."(P3)

3.2.2 Moral and Self-Sacrificing Police Identity. Officers often internalize a moral and self-sacrificing image of their profession, seeing their role as a calling and taking great pride in contributing to society. As P7 noted, "Police officers throw themselves into their profession". This strong sense of purpose, however, often clashes with the organizational reality of efficiency-driven expectations. The pressure to prioritize speed over empathy creates emotional conflict, as officers struggle to reconcile their moral ideals with the practical demands of their work. Adding to this tension, officers feel constantly scrutinized by the public and held to high moral standards:

"As the police uniform symbolizes, we are always in the spotlight and highly visible, making ours the first profession to be held accountable."(P3)

This perpetual scrutiny reinforces a professional identity that leaves little room for vulnerability, pushing officers to internalize their struggles and avoid seeking help. This fear drives many officers toward avoidance behaviors, such as resigning, transferring roles, or simply accepting their situation without seeking professional help. As one participant experiencing moderate to severe depression shared:

“Honestly, I don’t really know how to deal with these struggles. I just take each day as it comes and get through it however I can.”(P2)

Instead, colleagues often become the primary—and sometimes only—source of emotional support, fostering a strong sense of solidarity within teams. As one participant remarked, *“The only people we can trust are our teammates”*(P12). Another added, *“Just by looking into each other’s eyes or at the case system, we can immediately tell what’s difficult”*(P3).

While this close-knit team culture provides emotional support, it also creates additional burdens. Officers may hesitate to use formal support systems, such as parental leave, fearing that doing so might increase their colleagues’ workload. This normalization of mental health struggles as an inevitable part of their profession often prevents officers from seeking formal help. Over time, such normalization can lead to profound disillusionment, especially as they witness the physical or psychological toll their work takes on their peers.

Despite these challenges, many officers continue to endure their struggles in silence. This tendency to endure is closely tied to the perceived expectations of their professional image, where acknowledging vulnerability clashes with the ideal of a steadfast and reliable officer. One key barrier is the fear of potential career repercussions. With promotions often tied to peer evaluations, officers worry that utilizing workplace counseling services could label them as weak officers, potentially impacting their career progression. In cases of severe distress, such as suicidal ideation or panic attacks, one officer sought external services to ensure privacy. They described their journey to seek help after experiencing intense verbal abuse from a complainant, which triggered panic symptoms.:

“When I started having thoughts of wanting to die... I decided to seek out a hospital. As for the counseling services offered by the company... I know they’re good institutions, but I worry about running into coworkers at those facilities or the counseling center, and what might come up as a result.”(P1)

3.3 Routine Exposure to Negative Events

Korean police officers frequently confront the darker facets of society, often in ways that are deeply intertwined with their everyday reality. These encounters require officers to make complex value judgments while maintaining emotional detachment, a process that can be both mentally and emotionally taxing. The repeated exposure to human tragedy and moral ambiguity not only challenges their professional resilience but also fosters a growing sense of fear and distrust toward the world they navigate.

3.3.1 Mediator Role in Distressing Incidents. Police officers frequently face the heavy burden of making critical value judgments

in high-stakes situations, where their decisions can have profound consequences for others. One participant described the mental strain of balancing empathy and accuracy in these situations:

“It’s not just about stacking objects... We have to assess the situation at the time of the incident, listen to both parties, and, if necessary, even empathize with them... and we must not make mistakes.”(P3)

This meticulous approach is essential, yet it places a heavy emotional and cognitive load on officers who must quickly adapt to each case. The challenge is compounded by the need to understand the psychological nuances of strangers, requiring officers to adjust their investigative approach dynamically based on behavioral cues.

Moreover, the nature of police work often involves delving deeply into the negative and private aspects of others’ lives. This can create discomfort, as officers must navigate blurred boundaries between professional investigation and personal intrusion, while also enduring public scrutiny. Such interactions often provoke moral distress, especially when officers empathize with victims’ pain but are forced to process these emotions as part of their job. The constant exposure to intense emotional appeals, such as threats of self-harm from victims, further intensifies this burden:

“Fraud victims often say things like, ‘I can’t take this anymore, I’m going to end my life—please hurry.’ ... It makes me wonder, ‘Did this happen because I didn’t handle it well enough?’ Thinking like that... I think that’s how I ended up feeling so stressed.”(P1)

The combination of overwhelming caseloads and the need to address severe cases quickly often results in officers feeling morally compromised, as they may view urgent issues as mere tasks to be completed.

3.3.2 Emotional Resilience Tested by Human Tragedy. Continuous exposure to the darker side of humanity—fraud, deceit, and even death—leads many officers to develop a sense of distrust and fear toward the world, often resulting in small-t traumas. Small-t trauma or micro-trauma refers to repeated exposure to distressing events that, while not life-threatening, accumulate over time and leave a significant psychological impact[7, 31]. These cumulative experiences profoundly shape their worldview:

“So this is how people deceive others in the world... There really are so many people who are mentally unwell. ... I end up directly encountering the kind of terrible things that I wouldn’t need to know about, things I’d normally only see on the news—if I weren’t a police officer.” (P2)

Despite the emotional toll, the nature of their duties demands that officers remain emotionally detached, leaving little room for vulnerability or expressions of fear. This requirement for emotional detachment exacerbates feelings of isolation and desensitization, making it increasingly difficult for officers to process their experiences in a healthy and constructive way.

Furthermore, while frequent exposure to individuals with mental health challenges might typically reduce stigma, the nature of the cases officers encounter often has the opposite effect[17, 29]. Repeatedly interacting with people in severe crises or engaging in harmful behaviors reinforces negative perceptions of mental illness.

This, in turn, influences officers' help-seeking attitudes, making them less likely to seek mental health support themselves and deepening their emotional isolation. This stigma is compounded by a lack of confidence in external support systems.

In addition, a 'Perceived Lack of External Empathy' prevents others from truly understanding their struggles. Officers often feel that their unique experiences and internal dynamics are too complex to explain or be understood by outsiders, including family members:

"Honestly when I try to share it with my family, they just don't seem to fully understand or empathize with this part of my experience."(P3)

3.4 Misalignment Between Internal Growth Needs and External Acknowledgement

While a strong sense of duty and dedication serves as the foundation for police officers' commitment to their profession, these very values are often undermined by external pressures, societal expectations, and rigid organizational practices. Despite their efforts to uphold justice and assist others, officers frequently find their internal growth needs unmet and their contributions overshadowed by an overemphasis on quantitative assessments. This misalignment erodes their sense of accomplishment, replacing fulfillment with frustration and disillusionment.

3.4.1 Public Expectations and Emotional Burdens. In South Korea, police officers are metaphorically referred to as the "people's cane", symbolizing their role as public servants expected to comprehensively resolve societal issues. This perception places immense emotional and psychological demands on officers, as they are often treated as problem-solvers for all public grievances:

"Because citizens think of us as part of a legal service industry now."(P3)

Dealing with negative attitudes from the public further compounds their stress. According to interviews, a single intense interaction with a difficult complainant can leave a lasting emotional impact, resulting in chronic tension and anxiety, and, in some cases, even leading to the onset of panic disorder. Moreover, police officers are often perceived not as community supporters but as individuals only encountered when something bad happens. This stigma portrays them as hostile figures to be avoided, further intensifying their emotional burdens:

"There are many cases where people view the police with hostility, almost with hatred."(P6)

3.4.2 Task Significance Overshadowed by Quantitative Metrics. The overemphasis on quantitative evaluations diminishes the intrinsic value of police work and undermines officers' morale. While some officers find fulfillment in resolving cases and experiencing a sense of accomplishment—statistical systems used to measure performance often fail to capture the qualitative significance of their responsibilities. This narrow focus on numerical metrics instead leads to frustration and a sense of futility:

"Even when there's a lack of manpower, instead of acknowledging that we've been working hard, it's just, 'You're not doing well enough, so work harder.' ... We end up being judged harshly based solely on numbers,

without any consideration for our effort or the challenges we face."(P12)

Such systems not only negate the subjective interpretation of a case's importance but also erode officers' sense of accomplishment. As one participant expressed:

"No matter how hard I work, it feels like I'm not improving at all. I started losing confidence in myself."(P2)

This disconnects between the officers' internal drive for professional growth and the external emphasis on numerical assessments perpetuates feelings of disillusionment, diminishing their overall well-being.

4 Discussion

Research on the mental health of police officers often focuses on general stressors or high-level interventions, with limited attention to the nuanced needs of specific roles like investigative officers. The aim of this study was to derive design implications for DMHIs for Korean investigative police officers by exploring their lived experiences and challenges. The findings emphasize that the demanding and unpredictable nature of investigative work, combined with cultural and organizational barriers, significantly affects officers' mental health. This aligns with existing literature, which underscores the need for interventions that address both systemic and individual factors[8, 15, 33]. To meet this need, the following design implications were derived based on the interview themes.

4.1 Integrating Micro-Intervention Programs into Police Work

Investigative police officers face significant mental health challenges due to demanding workloads and unpredictable tasks with tight deadlines, leaving little room for voluntary help-seeking[19]. Officers emphasize the need for formalized and mandatory interventions that are seamlessly integrated into their schedules, ensuring accessibility without adding to their workload pressures. These interventions must be proactive and designed to fit within the constraints of their professional routines. Mental health support should take the form of brief, actionable micro-interventions that are embedded into daily workflows. These interventions should be comparable to annual health checkups but tailored to address the unique stressors of police work. Role-specific mental health surveys can be used to flag red flags, indicating significant risks. The flagged issues must then prompt immediate follow-ups with offline services, with institutional guarantees for time-off to ensure officers can participate without additional workload burdens.

To ensure timely delivery, Just-in-Time Adaptive Interventions (JITAI) can serve as a framework to predict optimal moments for intervention[34]. Using real-time data from digital phenotyping and periodic mental health surveys[6, 30], future research should explore how JITAI can identify high-risk situations and determine the optimal moments when officers are most likely to benefit from targeted support. These interventions must be micro in nature, designed to be minimally intrusive and tailored to align with officers' roles, workloads, and stress levels. The interventions themselves should be compact and actionable. For instance, new recruits

might receive motivational messages or resilience-building strategies, while senior officers managing high-stakes cases could benefit from techniques for stress management or conflict resolution. Routine workflows, such as handling complaints, could integrate these interventions, combining immediate psychological relief with professional insights. Sharing senior officers' advice, paired with expert interpretations, can further enrich the interventions and provide practical tools for managing stress. By embedding JITAI-guided micro-interventions into police workflows, this approach shifts mental health care from a voluntary and reactive model to a structured and proactive one.

4.2 Recognizing The Thriving Self with a Sense of Purpose and Pride

Korean investigative police officers face relentless negativity in their work, including hostile interactions, public criticism, and media scrutiny. This is particularly pronounced in South Korea, where cultural expectations of police officers as moral and self-sacrificing figures amplify the burden of public and media judgment. These societal pressures, combined with a lack of positive reinforcement, contribute to emotional exhaustion, reduced job satisfaction, and burnout. Despite these challenges, their strong sense of duty and responsibility often serves as the driving force that keeps them committed to their work, even in the face of overwhelming negativity. To counter this imbalance, it is critical to rethink how police officers' efforts and contributions are evaluated and rewarded. Many officers define their workload not just by the quantity of tasks but by the intensity and emotional labor required. Therefore, aligning positive rewards with these internal growth needs—such as recognition for perseverance, decision-making in crises, or emotional resilience—can foster a more meaningful sense of accomplishment. Positive psychology offers a valuable framework for achieving this by shifting the focus from numerical metrics to acknowledging the intangible but vital aspects of their work[10, 24].

Brief and frequent mechanisms—such as personalized recognition, micro-celebrations, or public gratitude dashboards—can counteract chronic exposure to criticism while fostering resilience and intrinsic motivation. Tools that quantify and visualize the cumulative impact of officers' work, such as dashboards showing lives impacted or safety improvements, provide a way to align their efforts with meaningful outcomes, ensuring their contributions are acknowledged and valued. A structured positive reward system tailored to the cultural context of South Korea can transform the emotional landscape of policing. By promoting resilience, reducing burnout, and enhancing job satisfaction, this approach strengthens officers' sense of purpose while addressing the imbalance of negative and positive feedback. Furthermore, it acknowledges the emotional labor intrinsic to police work and positions it not just as a duty, but as a deeply valued and appreciated contribution to society.

4.3 Toward Improving Mental Health Literacy

The culture of “enduring in silence” among Korean investigative police officers stems from stigma, fear of vulnerability, and the belief that outsiders cannot understand their struggles. While officers rely heavily on colleagues for emotional support, this reliance is

often informal and insufficient. Developing an anonymous digital communication platform exclusively for colleagues can facilitate a cultural shift toward “mutual reliance,” emphasizing that “it’s okay to lean on others” and normalizing open discussions about mental health. This digital space could provide a safe environment for officers to share their struggles and receive peer support, leveraging the strong camaraderie they already share while reducing barriers to open communication[11].

However, the effectiveness of such a platform depends on addressing key challenges, such as lowering entry barriers for initial engagement and ensuring long-term retention of participants[4]. Officers' low mental health literacy and hesitation to openly discuss such topics could limit participation. To mitigate this, the platform must integrate structured professional guidance and curated mental health resources, ensuring it serves as both a supportive space for peers and a reliable source of expert-driven insights. For example, moderated discussions, guided activities, and access to practical mental health tools could bridge the gap between officers' needs and their willingness to engage. Community-building tools, such as anonymous peer-to-peer letters or emotional feedback dashboards, can further encourage participation without demanding significant time or effort. These tools not only foster connection but also allow officers to recognize shared challenges, reducing feelings of isolation and stigma. Embedding mutual reliance into police culture through a professionally supported digital platform not only reduces stigma but also fosters resilience and a sense of belonging.

Future research should explore ways to optimize engagement, focusing on reducing entry barriers and enhancing long-term retention within such platforms. Feedback mechanisms, such as peer reviews of counseling services or recognition for meaningful contributions to the platform, could help sustain participation by linking engagement to positive reinforcement. By combining expert-driven support with peer-based interaction, this approach creates a more cohesive and supportive workplace culture, equipping officers with both the knowledge and connections they need to navigate mental health challenges effectively.

5 Conclusion

This study explores the occupational and cultural contexts of investigative police officers and proposes design implications for DMHIs tailored to their unique mental health challenges. Existing digital mental health solutions for police officers remain underdeveloped and lack context-specific nuance. By identifying key stressors and intervention strategies, this work offers a foundation for developing role-specific and culturally sensitive mental health support. While focused on investigative officers, these insights highlight the need to extend digital interventions to other police departments with distinct occupational demands.

Future research should examine role-specific stressors in patrol, forensic, cybercrime, and administrative units to ensure broader applicability. Additionally, as mental health attitudes are shaped by cultural contexts, cross-national studies are needed to refine interventions for different policing systems. Effective implementation of these interventions requires consideration of organizational structures and systemic barriers. The hierarchical nature of police organizations may pose challenges in adoption, emphasizing the

need for leadership engagement, formal resource integration, and structural incentives. Addressing these factors will be key to ensuring sustainable mental health support within policing and other high-stress professions.

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